Si calcolino i seguenti integrali indefiniti:

(1) $\int x^2 e^{x^3+1}\,dx$;

(2) $\int \frac{1}{x}\log^2 x\,dx$;

(3) $\int \cos^2 x \sin x\,dx$;

(4) $\int \frac{x^2}{1+x^3}\,dx$;

(5) $\int \frac{e^x}{e^{2x}+1}\,dx$;

(6) $\int (x^2+1)e^x\,dx$;

(7) $\int \cos^2 x\,dx$;

(8) $\int x^2 \cos x\,dx$;

(9) $\int x \log^2 x\,dx$;

(10) $\int x^3 \log x\,dx$;

(11) $\int \frac{x-1}{x^2-4}\,dx$;

(12) $\int \frac{2x}{x^2-3x+2}\,dx$;

(13) $\int \frac{x+1}{(x-3)^2}\,dx$;

(14) $\int \frac{2x+1}{x^2+4}\,dx$;

(15) $\int \frac{x+3}{x^2+9}\,dx$.

Si calcolino i seguenti integrali definiti:

$$\int_0^{\pi/2} x^3 \cos x\,dx; \qquad \int_0^3 x^2 e^x\,dx;$$
$$\int_1^2 \frac{x-2}{x^2-9}\,dx; \qquad \int_0^2 \frac{x-1}{x^2+3}\,dx;$$
$$\int_2^3 \frac{x}{(x+2)^2}\,dx; \qquad \int_{1/e}^e \log x\,dx.$$